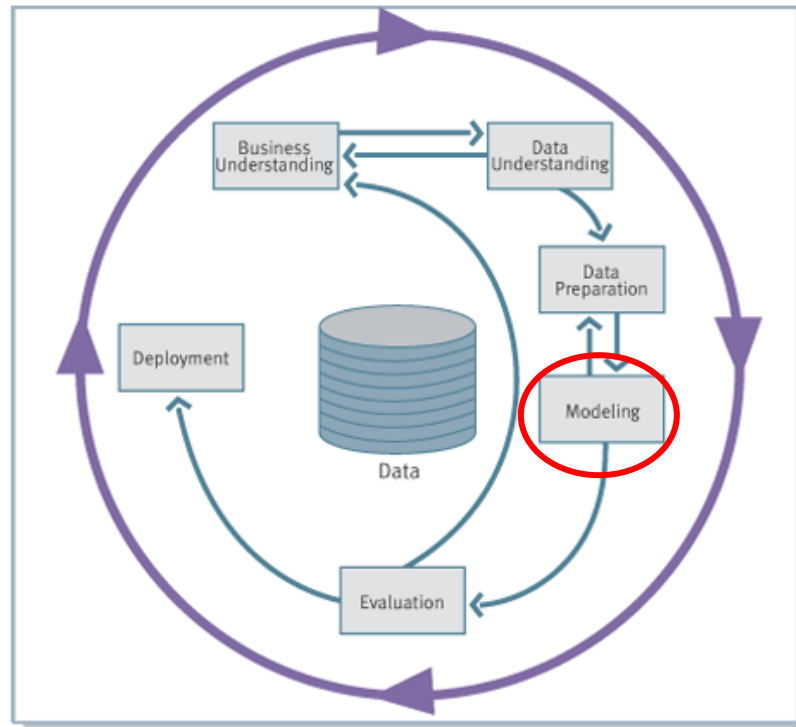


Modeling



Modelling

Select Modeling Technique

- The actual modeling technique is chosen among pre-selected techniques and tools.



OUTPUT

- the modeling technique and any modeling assumption by this technique.

Examples of assumptions: probability distribution, no missing values, class attribute is nominal, etc.

Modelling

Select Modeling Technique

The data analyst will select the specific algorithm(s) to run on the basis of a set of factors, including:

- the nature of the objective (classification, regression, etc.);
- ability to handle certain data types;
- ability to handle multiple relations and to generate relational patterns;
- explicability of the output;
- scalability;
- level of familiarity.

Modelling

Select Modeling Technique

One can identify three primary components in any data mining algorithm:

- **model representation**
 - Relational vs. propositional, quantitative vs. qualitative, appropriateness for the intended user (human, computer program, data mining algorithm)
- **model evaluation**
 - Model uncertainty is based on probability estimates, statistical tests, fuzzy theory, etc.
- **search**
 - (exhaustive, greedy, ...)

The selection of the algorithm(s) is also based on these factors.

Data Mining Algorithm

Model representation

Model representation is the language L used to describe discoverable patterns. It depends on:

- the **type** of the representation,
- the **descriptive capacity**,
- its **appropriateness** for the intended user.

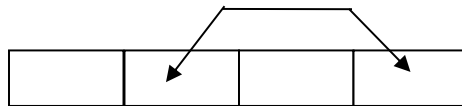
Data Mining Algorithm

Model representation

Discovered knowledge can be categorized by the **type** of data pattern.

Data pattern may be either **interfield patterns** or **interfield-record patterns**.

Interfield patterns relate values of fields in the same record



Procedure = surgery $\Rightarrow$ days-in-hospital > 5

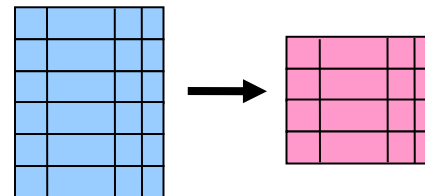
Data Mining Algorithm

Model representation

Interfield-record patterns relate values aggregated over groups of records

Example:

Diabetic patients have twice as many complications as non-diabetics



Data Mining Algorithm

Model representation

Descriptive capacity:

- A **quantitative discovery** relates numeric field values using mathematical equations.

$$X^2 + 0.5 * Y = 2.65$$

- A **qualitative discovery** finds a logical relationship among fields.

Procedure = surgery => days-in-hospital > 5

Data Mining Algorithm

Model representation

Quantitative and qualitative discoveries are often expressed as **simple rules**.

*If the procedure is surgery **than** days-in-hospital are greater than 5*

Putting together several simple implications we obtain a logical theory that, under some assumptions, can be considered a causal theory.

Data Mining Algorithm

Model representation

Representation should be appropriate for the intended user.

The users may be:

- Humans
- Computer programs,
- The discovery system itself.

Data Mining Algorithm

Model representation

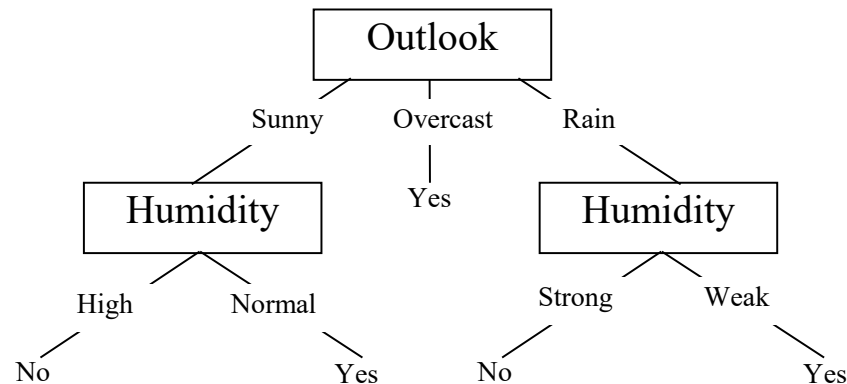
- The users are Humans
 - **Natural language**: desirable, but not convenient for manipulation by discovery algorithms.
 - **Logical**: more natural for computation and, if necessary, can be translated into a natural language form.

Data Mining Algorithm

Model representation

Examples:

- Decision trees



- Logical representation

IF Parent(x,y) THEN Ancestor(x,y)

IF Parent(x,z) AND Ancestor(z,y) THEN Ancestor(x,y)

These representational forms offer different advantages and limitations.

Data Mining Algorithm

Model representation

- **Visual depictions**: The relationships among the branches of a decision tree are more evident in a graphic presentation than as a logical representation of nested conditional statements

IF Outlook = sunny THEN

 IF Humidity = high THEN PlayTennis = No

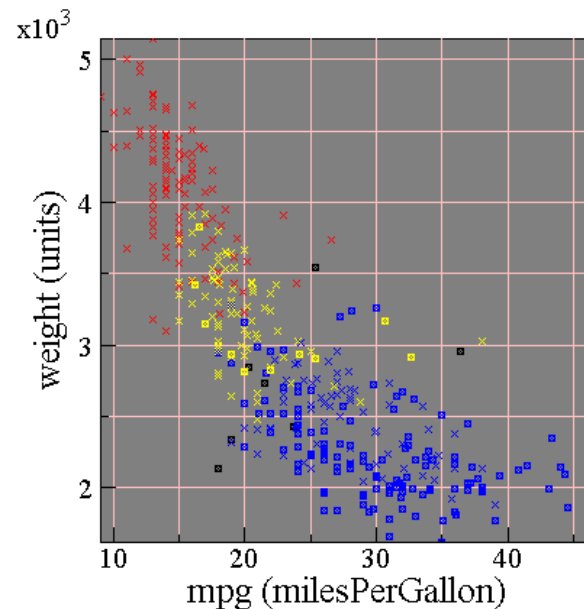
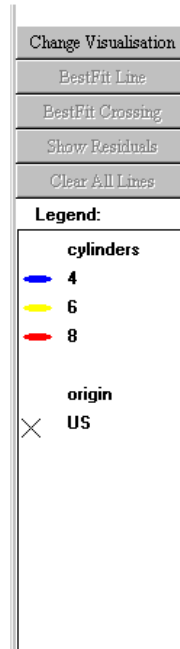
 ELSE IF Humidity = normal THEN PlayTennis = Yes

ELSE ...

Data Mining Algorithm

Model representation

Information about the **shape and density** of clusters of records is another type of knowledge that is best presented **visually** by means of two- or three-dimensional plots Visual depictions



Data Mining Algorithm

Model representation

- The users are **computer programs**, such as expert systems shells; in this case, appropriate representations include **programming languages** and **declarative formalisms** (e.g. Prolog programs).
- The user is **the discovery system** itself: when discoveries are fed back into the system as domain knowledge, domain knowledge and discovered knowledge must share a common representation.

Data Mining Algorithm

Model representation: a critical choice

- If the representation is too limited, then no amount of training time or examples will produce an accurate model for the data.
- It is important that a data analyst fully comprehend the representational assumptions that may be inherent to a particular method

Data Mining Algorithm

Model representation: a critical choice

Warning:

A more powerful representation formalism for models increases the danger of **overfitting** the training data resulting in **reduced prediction accuracy on unseen data**.

Data Mining Algorithm

Model representation: uncertainty

Patterns are often associated with a **degree of uncertainty**.

Methods to represent uncertainty are:

- Probability,
- Standard deviation,
- Fuzzy sets.

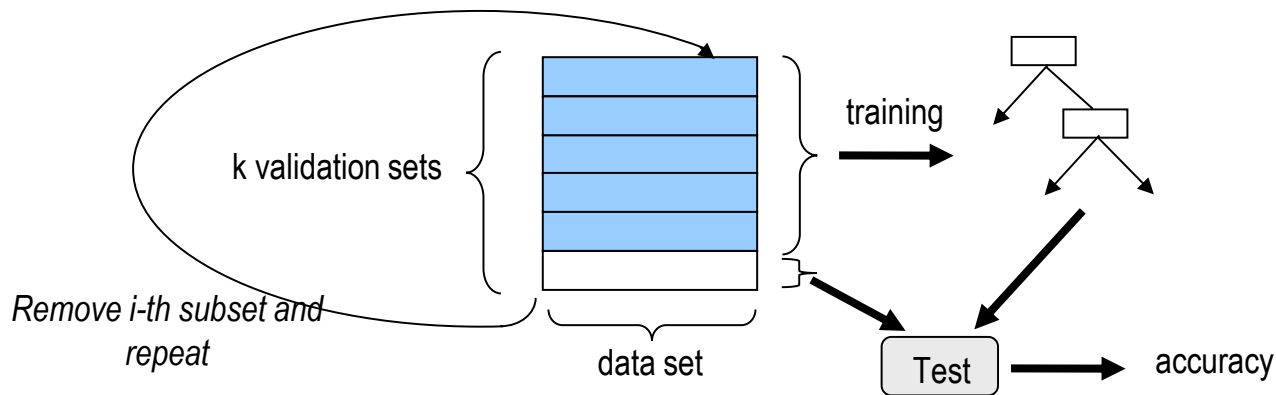
Data Mining Algorithm

Model evaluation

Model evaluation estimates the uncertainty of a particular pattern.

There are many ways to evaluate this uncertainty.

Cross validation is one of them.



Data Mining Algorithm

Model evaluation

Cross Validation procedure:

1. Shuffle the items in the data set 1.
2. Divide the data set into k equal subsets of size n .
3. For $i := 1$ to k do
 - a) Call the i -th subset of n items the *test set* and put it aside.
 - b) Train the system on the remaining $k-1$ subsets.
 - c) Test the system on the test set and record the performances.
4. Clear memory, i.e., forget everything learned during training
5. Calculate the average performance from the k test sets

Modeling

Generate Test Design

Before building models, it is important to generate a procedure to test the quality of models.

Example:

If the data mining task is classification, it is common to use error rates as quality measures for data mining models. Therefore the data is separated into a training set and a testing set.

OUTPUT:

- The test design, which describes the intended plan for training, testing and evaluating models.

Modeling

Build Model

The modeling (i.e., data mining) technique is applied to the data.

OUTPUT:

- Parameter settings and the rationale of their choice
- Output models together with expected accuracy, robustness, and possible shortcomings.

Build Model

Output in PMML

The output of a data mining algorithm can be expressed according to some industrial standards.

The **Predictive Model Markup Language (PMML)** is an XML-based standard supported by

- Full Members: IBM, Oracle, Magnify, SPSS, SAS, StatSoft, Microsoft, CorporateIntellect, KXEN, Salford Systems
- Numerous Associated Members
- <http://www.dmg.org/>
- Objective
 - define and share predictive models using an open standard

Build Model

Output in PMML

- Complex mosaic of software applications
 - Knowledge generators
 - Data Mining Vendors
 - Different data mining algorithms have different languages for expressing the knowledge discovered
 - Vendor dependent representations for knowledge e.g. C/C++ routines
 - Knowledge consumers
 - Real-time Scoring / Personalisation engines
 - Marketing Tools
 - Visualisation Tools
- Need for a vendor independent representation of data mining output

Build Model

Output in PMML

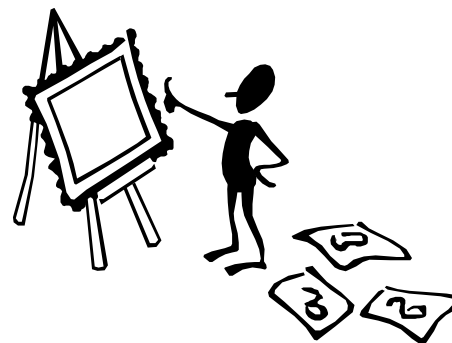
- Benefits
 - proprietary issues and incompatibilities no longer a barrier to the exchange of models between applications
 - develop models using any generator vendor, deploy the models using any consumer vendor application.
- Development
 - PMML Version 4.4 was approved in November, 2019
 - Supported by most current releases of member vendors applications

Build Model

Output in PMML

Supported **data mining models**

- Tree Model
- Neural Networks
- Clustering Model
- Regression Model
- General Regression Model
- Naïve Bayes Model
- Association Rules
- Sequence Rule Model
- Text Model (v. 3.1)
- Rule Set (v. 3.1)
- Support Vector Machine (v. 3.1)



Modeling

Assess Model

Modeling results are interpreted according to data mining success criteria and the desired test design. A model assessment should answer questions such as:

- How accurate is the model?
- How well the model describe the observed data?
- How much confidence can be placed in the model's predictions?
- How comprehensible is the model?

The answer depends on the type of model that was built.

OUTPUT:

- Assessment of generated models
- Revision of parameter settings

Modeling

Assess Model

Assessing descriptive models.

The rule:

if (state = 'MA') then heating source = oil

seems more descriptive than the rule:

if (area=339 or area=352 or area=413 or area=519) then heating source = oil

Even if the two rules turn out to be equivalent, the first one seems more expressive. Expressive power may seem purely subjective, but there is, in fact, a theoretical way to measure it. *The minimum description length* (MDL) for a model is the number of bits it takes to encode both the rule and the list of all exceptions to the rule. The fewer bits required, the better the rule.

Modeling

Assess Model

Assessing predictive models. Predictive models are assessed on their accuracy on previously unseen data.

- Model assessment can take place at the level of the whole model or at the level of individual predictions.
- Two models with the same overall accuracy may have quite different levels of variance among the individual predictions.
- Model assessment should be based on a test set independent from the training set.

Modeling

Assess Model

- For classification models, the **accuracy** estimate is the overall number of correct classifications, divided by the total number of samples in the testing set. The **misclassification** (or **error**) **rate** is the complement of the accuracy estimate.
- Accuracy may not be appropriate when errors may have different costs (and consequences).
- In that case an **error cost** is a better measure than a misclassification error.

Modeling

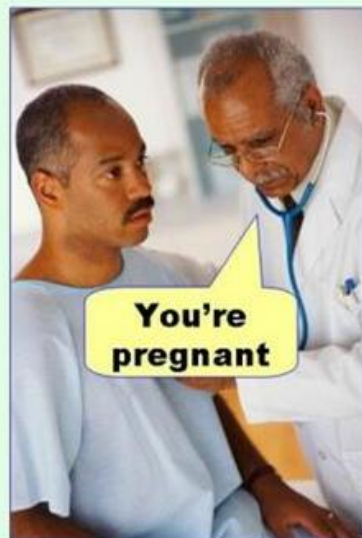
Assess Model

- We can use the **confusion matrix** to assess the performance of a classifier.

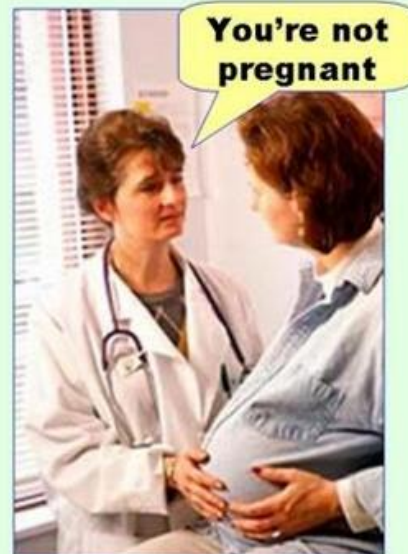
	Actually Positive (1)	Actually Negative (0)
Predicted Positive (1)	True Positives (TPs)	False Positives (FPs)
Predicted Negative (0)	False Negatives (FNs)	True Negatives (TNs)

Modeling Assess Model

(false positive)



(false negative)



Modeling

Assess Model - Accuracy

The most intuitive metric, accuracy is the percentage of correctly classified instances out of the total number of samples.

$$Accuracy = \frac{TP + TN}{TP + TN + FP + FN}$$

- Easy to understand: It has a straightforward interpretation.
- Sensitive to class imbalance: In highly imbalanced scenarios (e.g., fraud detection, where the majority of cases are not fraud), a model can achieve deceptively high accuracy by simply predicting the majority class.

Modeling

Assess Model - Precision

Proportion of predicted positive cases that are truly positive

$$\text{Precision} = \frac{TP}{TP + FP}$$

- Crucial when false positives are costly as in:
 - Medical diagnosis (avoiding unnecessary treatments)
 - Spam filtering (preventing misclassification of legitimate emails)
 - Fraud detection (avoiding flagging legitimate transactions)

Modeling

Assess Model - Recall

Proportion of actual positive cases correctly identified by the model

$$\text{Recall} = \frac{\text{TP}}{\text{TP} + \text{FN}}$$

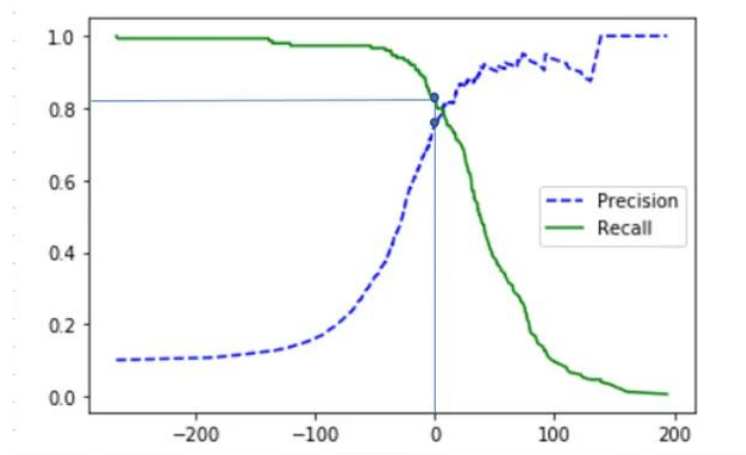
- Crucial when false negatives are costly as in:
 - Security systems (detecting all potential intrusions)
 - Manufacturing defect detection (identifying all faulty products)

Modeling

Assess Model – Precision vs Recall

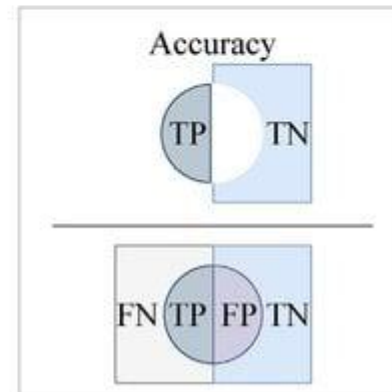
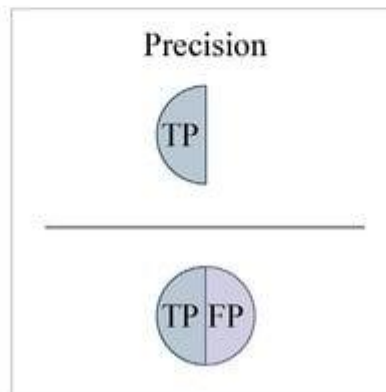
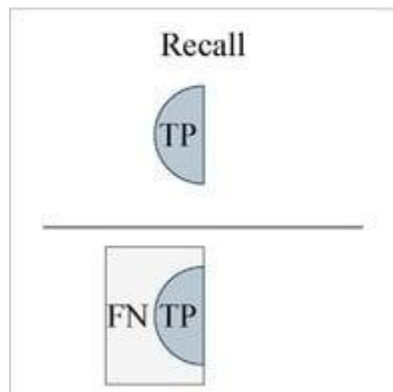
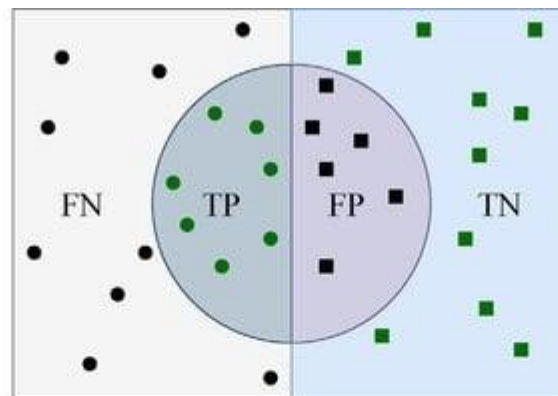
Unfortunately, you can't have both precision and recall high. If you increase precision, it will reduce recall, and vice versa.

This is called the **precision/recall tradeoff**.



Modeling

Assess Model



Modeling

Assess Model – F1 Score

F1 Score is a metric for classification problems that combines precision and recall.

It's the harmonic mean of precision and recall values.

$$F1 = 2 \times \frac{\text{Precision} \times \text{Recall}}{\text{Precision} + \text{Recall}}$$

- Used when we want to balance both precision and recall in our model evaluation.
- Harmonic mean is used instead of arithmetic mean because it penalizes extreme values more effectively.

Modeling

Assess Model – Example

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: ?

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Modeling

Assess Model – Example

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: 165

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: 165

Classifier Predictions: ?

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: 165

Classifier Predictions:

- 110 Positive,
- 55 Negative

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: 165

Classifier Predictions:

- 110 Positive,
- 55 Negative

Actual cases: ?

Modeling

Assess Model – Example

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ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: 165

Classifier Predictions:

- 110 Positive,
- 55 Negative

Actual cases:

- 105 disease-positive patients
- 60 disease-negative patients

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
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Predicted classes:

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Total predictions: 165

Classifier Predictions:

- 110 Positive,
- 55 Negative

Actual cases:

- 105 disease-positive patients
- 60 disease-negative patients

Accuracy (A): Overall, how often is the classifier correct?

$$A = (TP+TN)/\text{Total}$$

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
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Total predictions: 165

Classifier Predictions:

- 110 Positive,
- 55 Negative

Actual cases:

- 105 disease-positive patients
- 60 disease-negative patients

Accuracy (A): Overall, how often is the classifier correct?

$$A = (TP+TN)/Total = \mathbf{(100+50)/165 = 0.9}$$

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Predicted classes:

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- Negative (disease absent)

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Accuracy (A): Overall, how often is the classifier correct?

$$A = (TP+TN)/Total = (100+50)/165 = 0.9$$

Precision (P): When it predicts yes, how often is it correct?

$$P = TP / (TP + FP)$$

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
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- 105 disease-positive patients
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Accuracy (A): Overall, how often is the classifier correct?

$$A = (TP+TN)/Total = (100+50)/165 = 0.9$$

Precision (P): When it predicts yes, how often is it correct?

$$P = TP / (TP+ FP) = 100/110 = 0.91$$

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Predicted classes:

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- 60 disease-negative patients

Accuracy (A): Overall, how often is the classifier correct?

$$A = (TP+TN)/Total = (100+50)/165 = 0.9$$

Precision (P): When it predicts yes, how often is it correct?

$$P = TP / (TP+ FP) = 100/110 = 0.91$$

Recall (R): When it's actually yes, how often does it predict yes?

$$R = TP / (TP+ FN)$$

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

Predicted classes:

- Positive (disease present)
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Accuracy (A): Overall, how often is the classifier correct?

$$A = (TP+TN)/Total = (100+50)/165 = 0.9$$

Precision (P): When it predicts yes, how often is it correct?

$$P = TP / (TP+ FP) = 100/110 = 0.91$$

Recall (R): When it's actually yes, how often does it predict yes?

$$R = TP / (TP+ FN) = 100/105 = 0.95$$

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

F1-Score (F1): What is the weighted average of recall and precision?

$$F1 = 2 * (P * R) / (P + R)$$

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: 165

Classifier Predictions:

- 110 Positive,
- 55 Negative

Actual cases:

- 105 disease-positive patients
- 60 disease-negative patients

Accuracy (A): Overall, how often is the classifier correct?

$$A = (TP+TN)/Total = (100+50)/165 = 0.9$$

Precision (P): When it predicts yes, how often is it correct?

$$P = TP / (TP + FP) = 100/110 = 0.91$$

Recall (R): When it's actually yes, how often does it predict yes?

$$R = TP / (TP + FN) = 100/105 = 0.95$$

Modeling

Assess Model – Example

	PREDICTED: YES	PREDICTED: NO
ACTUAL: YES	100 (TP)	5 (FN)
ACTUAL: NO	10 (FP)	50 (TN)

F1-Score (F1): What is the weighted average of recall and precision?

$$F1 = 2 * (P * R) / (P + R) = 2 * (0.91 * 0.95) / (0.91 + 0.95) = 0.9296$$

Predicted classes:

- Positive (disease present)
- Negative (disease absent)

Total predictions: 165

Classifier Predictions:

- 110 Positive,
- 55 Negative

Actual cases:

- 105 disease-positive patients
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$$A = (TP+TN)/Total = (100+50)/165 = 0.9$$

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$$P = TP / (TP + FP) = 100/110 = 0.91$$

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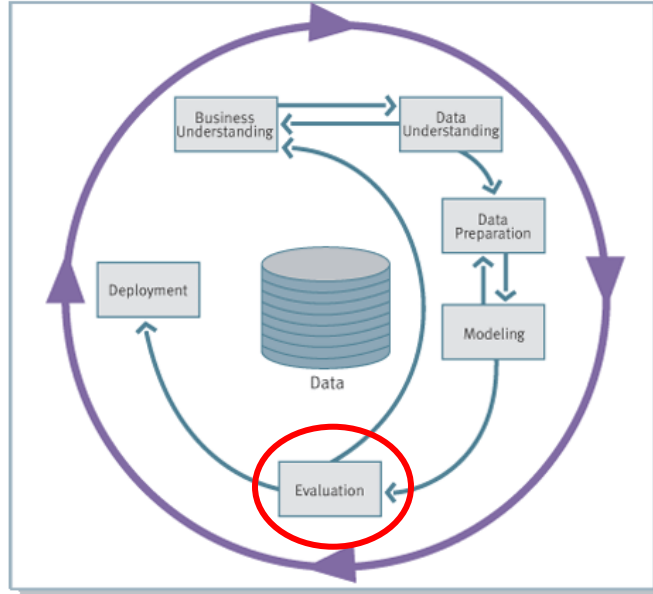
Modeling

Assess Model

- For regression models or for parameter estimation models, the accuracy is estimated as the **average sum of square errors** (differences between predicted and actual values) (mean square error).

$$\text{MSE} = \frac{1}{n} \sum_{i=1}^n (y_i - \hat{y}_i)^2$$

Evaluation



Evaluation

Evaluate results

Whereas model assessment deals with factors such as accuracy and generality of models, this task evaluates models according to the original business objectives and success criteria. The challenge is to present the new findings in a convincing, business-oriented way. This task can be properly done by a **skilled data analyst** working **with a business analyst**.

OUTPUT:

- An overall assessment of data mining with respect to business success criteria, including a final statement as to whether the project already meets the initial business objectives.

Evaluation

Evaluate results



This step is within the domain of the **business analyst** and is focused on the executive sponsor. The experienced business analyst will be able to formulate the findings in a way that relates directly to the business objectives that were set for the project at the outset.

In addition, a **data analyst** with extensive domain and/or industry knowledge can be a great asset at this stage.



Evaluation

Evaluate results

The biggest benefit from the project may well be a recognition of, or a renewed focus on, **the importance of the corporate data asset** and **the power of data-driven solutions**.

This in turn can engender a culture within the organization that can promote larger, longer-term, **data-oriented initiatives** such as data warehousing and data marting.

Evaluation Review Process

In addition to model evaluation, the entire process is considered in order to determine if there is any important factor or task that is neglected or to identify a generic procedure to generate similar models in the future.

OUTPUT:

Recommendation for further activities.

Evaluation

Review Process

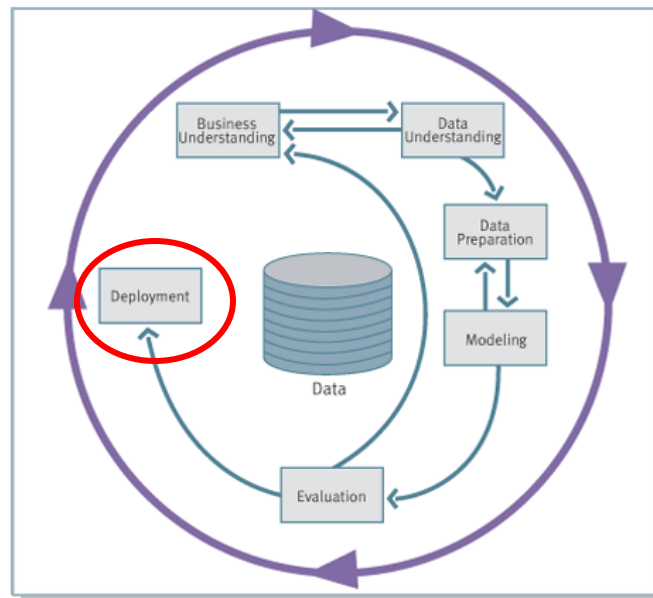
The key question is:

"Have we found something that is interesting, valid, and operational?"

Almost always the answer is no, in which case the exploratory cycle repeats itself, after having reviewed the process itself.

The specific activities in this step depend very much on the kind of application that is being developed.

Deployment



Deployment

Plan Deployment

In order to deploy the data mining results into the business, this task develops a strategy for deployment.

OUTPUT:

A deployment plan.

<i>How do you usually deploy data mining results? (Choose all that apply): [95 voters]</i>	
<i>Publish research papers (37)</i>	38.9%
<i>Use findings to change business rules (42)</i>	44.2%
===	
<i>Deploy in production and ... (46)</i>	48.4%
<i>Use data mining tool for scoring (47)</i>	49.5%
<i>Convert model to SQL (20)</i>	21.1%
<i>Convert model to another language (16)</i>	16.8%
<i>Convert model to C or Java (16)</i>	16.8%
<i>Convert model to PMML (4)</i>	4.2%
===	
<i>Deploy in batch mode (48)</i>	50.5%
<i>Deploy in real-time mode (21)</i>	22.1%

Deployment

Plan Deployment

The objective now is to put into action the commitments made in the opening step, according to the (presumably) new, valid, and operational information from the previous process steps.

The ***assimilation of knowledge*** demands for the formulation of ways in which the new information can be best exploited.

The specific business actions to be taken depend on the kind of application being developed and the executive commitments in the opening step.

Deployment

Plan Deployment

Technical issues.

Integration into the existing technical infrastructure could involve the inclusion of new predictive models and association rules in existing application code, expert system shells, or database procedures.



Deployment

Plan Monitoring and Maintenance

Careful preparation of a maintenance strategy helps to avoid unnecessarily long periods of incorrect usage of data mining results.

OUTPUT:

A maintenance plan on the monitoring process.



Deployment

Produce Final Report

At the end of the project it is better to write up a final report, which can be:

- A summary of the project
- A final presentation of the data mining results

OUTPUT:

A report or a final presentation or both



Deployment

Review Project

Assess what went right and what went wrong, what was done well and what needs to be improved.

OUTPUT:

Experience documentation help to consolidate important experiences made for future engagements. The documentation may include:

- pitfalls
- misleading approaches
- hints for selecting the best-suited data mining technique in similar situations.

Acknowledgments

These slides are based on original materials by Prof. Donato Malerba and Prof. Michelangelo Ceci, from the Department of Computer Science, University of Bari "Aldo Moro".